

SaTC 2.0: RES: Relational Coverage-Guided Differential Fuzzing for Critical Software

1 Problem Statement, Overview, and Objectives

Differential testing has long been considered one of the most powerful approaches to ensure the security, reliability, and correctness of software, *when applicable*. Differential testing offers a unique solution to the difficulty of specifying what a software system should (and should not) do: in essence, it tests a function f by checking its behavior matches function g that is *an independent implementation of the same functionality*, and defines test failures as cases where $f(x) \neq g(x)$. The “functions” f and g in realistic settings are seldom actual code-level functions, of course, but more typically complex systems ranging from data-structure implementations to complex parsers, cryptographic protocols [1] and, especially, compilers [2] and file systems [3].

The ubiquity of differential approaches to testing compilers and file systems is no accident: these are two critical pieces of system infrastructure where genuinely independent implementations are often available, and standards such as ANSI C and POSIX make comparison reasonable, if not trivial; the same applies, though less frequently, to cryptographic algorithm and protocol implementations. Of course, since file systems may rest on different hardware substrates, and compilers may make different choices where a language like C allows for implementation defined behavior, in practice even in these best-case scenarios, the problem is usually not one of checking that $f(x) = g(x)$, but of checking that $O_f(f(I_f(x))) = O_g(g(I_g(x)))$, where O_f , O_g , I_f , and I_g are respectively, output and input abstraction functions for f and g , that allow for known differences in behavior and input spaces, and there are often also known inputs that violate the differential specification in a known and allowed way, making the full specification of the form $P(x) \Rightarrow O_f(f(I_f(x))) = O_g(g(I_g(x)))$.

Despite these complexities, differential testing is extremely attractive, especially in settings where inputs are automatically (and stochastically) generated, such as random testing and *fuzzing*. Indeed, arguably the fundamental limitation of modern fuzzing, despite its huge success in finding software vulnerabilities and bugs, is the typical weakness of fuzzing oracles. It is extremely hard to determine if a system is behaving correctly on an arbitrary, often extremely exotic, input. This leads to results such as the difference between functional tests and fuzzing harnesses for Bitcoin core, where equivalent, almost 100% branch and code coverage yields huge discrepancies in mutation detection, with functional tests detecting almost 80% of injected bugs while fuzzing, despite perhaps more aggressive exploration of the state space, detects only 12% of these bugs [4]. A differential specification makes it possible for fuzzing to detect subtle semantic errors, not just memory safety violations and assertion failures.

It seems likely that the importance of differential fuzzing will only grow in the future, for two reasons, both deriving from the use of LLMs and agentic coding. First, the primary reason differential fuzzing is seldom used in practice is that the cost of an independent implementation of a software system, even in security or safety critical settings, is usually prohibitive. This cost is especially high given that previous work has shown that “independent” implementations are seldom as independent (in terms of correlation of bugs) as desirable [5]. This means that differential testing via separate implementations of functionality is usually only performed in cases of extremely complex specifications implemented by radically separated teams (e.g., compilers and file systems) with orthogonal incentives to produce their own implementations, or for very simple specifications such as data structures. Code synthesis by AI models sidesteps this problem: once a specification that produces a single workable implementation is available, having a single model produce multiple implementations is often relatively cheap. Using multiple models or changing prompts to focus code generation on, for example, a faster implementation, an implementation with a smaller memory footprint, a more readable implementation, an implementation using an unusual algorithm, and so forth, seems likely to produce controlled divergence that may exceed that of human developers. If the development cost of multiple implementations is radically reduced, the advantages of differential specification are clear.

One pre-AI way to work around the cost of a second implementation, of course, is to use differential fuzzing in a *regression* setting, comparing a new version of software to an older, better-tested version. However, in practice, new versions of a software system are often introduced precisely because the functional interface

has changed in some way, and the work required to conceive, implement, and maintain a mapping between new and old versions (which will be less stable, as each new version tends to remove some parts of the current mapping, as the “old” version is now the version that was new, as well as add new changes) discourages use in this setting, especially as the bug correlation problem reduces the power of differential testing. Again, LLMs are probably well-suited to constructing and maintaining version mappings at low cost¹.

Second, differential approaches seem ideal for probing ambiguities in specifications presented to AIs, and AI misunderstandings: if multiple implementations are produced from a specification, and checked for equivalence, supervisors of AI-driven development can examine cases where the AI’s “view” of the software’s behavior is unstable, and can work to remove ambiguity and correct mistakes.

Differential fuzzing is, of course, already an important practice, particularly in security-critical code development. However, differential fuzzing as practiced now is often limited in effectiveness. All too often, the approach taken is to independently fuzz two implementations, then compare their behavior over the union of the produced corpuses of interesting inputs. While this can discover bugs, it is effectively throwing away most of the power of fuzzing, in two important ways. First, the implementations are only tested differentially for a small number of inputs, the final results of fuzzing runs on the two separate systems; this misses out on the opportunity to detect divergence and bugs on potentially millions or billions of executions. Second, the two independent corpuses are only derived from efforts to cover behaviors of the two systems independently; if an input does not produce novelty in terms of either system in isolation, but represents a unique *pair of coverage patterns* for a shared input, this method is simply incapable of identifying and exploiting such inputs. Such inputs, however, are, intuitively, the most promising in terms of detecting divergences in implementation behaviors.

With somewhat more effort, users of differential fuzzing can compile two implementations in such a way that a fuzzer can see the combined coverage of both systems, essentially a program path over the merged program `r1 = f(x); r2 = g(x); assert (r1==r2);`. This overcomes the above objections, but still in fact leaves much to be desired, in terms of taking advantage of the differential nature of the specification.

1.1 First-Class Differential Fuzzing

In computer science, fully leveraging an approach tends to require making it truly *first-class*, supported by language and analysis tools; without OO constructs or native-language threads, objects and concurrency are limited in applicability, and the recent jump in sophisticated usage of functional programming techniques in mainstream languages can essentially be traced to the widespread availability of real lambdas and closures, rather than weaker proxies such as object dispatch simulations of lambdas, or C’s function pointers.

Similarly, differential specification (arguably the most powerful and “usable” of all specification techniques) can only be fully exploited when *fuzzing algorithms* and *specification techniques* are differential-aware. This requires abstracting away from the specifics of a particular instance of differential testing/fuzzing. The most important previous work along this line, the NEZHA framework of Pesios et al. [6] offered a start on such an approach, and proved extremely successful (finding differences between implementations at a rate 20-50 times higher than previous approaches) in practice, but failed to truly offer up first-class differential fuzzing.

In order to understand the limits of current approaches, it is useful to think about a very simplified description of what a modern mutation-based fuzzer such as AFL++ [7], libFuzzer [8], or Google’s Centipede [9], does, at a high level that essentially applies to all fuzzers:

1. First, a fuzzer selects an input, a *seed* that it has already executed (possibly presented to it as part of an initial *seed corpus* of interesting inputs, e.g., valid media files when fuzzing a media parser), based on a *seed scheduling* algorithm that selects the most promising such inputs to work with. At this stage the fuzzer may also compute an *energy*: how much time to spend working on this particular seed (energy can also be modified once the fuzzer starts seeing results of working with the seed, but we are offering a simplified version of the process here).

¹We believe this will be especially true if LLMs have access to a high-quality DSL for constructing differential harnesses, which is one of the goals of this proposal.

2. Then, repeatedly, up until the limits set by *energy*, the fuzzer *mutates* the seed, making a randomly chosen change to the input; these changes can be simple byte-based mutations (often referred to as “havoc” operators) or more targeted and domain-specific, as in some compiler-fuzzing tools [10].
3. For each such modified variant of the seed, the fuzzer executes the mutant of the seed; if a crash is detected (in differential fuzzing, the executable should be set up so that divergence in behaviors triggers a crash) the input is preserved as representing a potential bug. If the input does not cause a crash, the fuzzer determines if the input triggers *novel* behavior. The definition of novelty can vary; some fuzzers only consider inputs that cover new branches or basic code blocks novel (this is libFuzzer’s default mode, for example); other fuzzers (AFL and family, for example) make use of a more complex edge-hit-count pattern approach that can approximate a form of (acyclic) path coverage [11]. If the input is novel, it is added to the set of seeds to schedule and spend energy on collected by the fuzzer, and the loop eventually returns to step 1, when energy for the current seed is exhausted. The identification of novel inputs is what lets the fuzzer slowly accumulate new interesting program behaviors and deeply explore the program, searching for bugs.

First, NEZHA’s fundamental contribution was making a tuple representation of coverage across compared methods, $\langle \text{cov}(f), \text{cov}(g) \rangle$. This is a huge advance on simply attempting to cover both implementations independently, then compare over interesting paths for either approach; it captures the importance of asymmetry and unusual *pairings* of coverage paths between compared implementations. However, this is in essence already done by AFL/AFL++ and other fuzzers, if the differential targets are compiled into a single instrumented binary, and NEZHA’s experimental evaluation unfortunately only compares to application of AFL to generate inputs based on a single implementation. NEZHA’s δ -based measure of input novelty is essentially based on a flattening of output (in black-box settings) or program path (in grey-box settings) tuples, so that novel pairings of f and g behaviors are observable; when a single executable with a global path is produced from the differential fuzzing targets in AFL/AFL++, the same result is essentially achieved. The advantage NEZHA shows over an implementation of what they call “global coverage” in libFuzzer is due to libFuzzer’s simpler novel branch/block heuristic for determining which inputs are interesting.

However, despite missing the fact that their grey-box approach to δ -based novelty is likely simply equivalent to a single-compiled-binary AFL/AFL++ approach, the NEZHA paper shows how to *take differential fuzzing seriously*. This proposal aims to go beyond this approach by modifying fuzzer internals (implemented on top of the industry-standard AFL++ platform) to take advantage of the differential problem in all stages of the basic fuzzing algorithm presented above:

- First, and most important, we aim to extend the notion of novelty to preserve, and thus explore from, inputs that do not find novel behavior of either f or g alone, but which (going beyond NEZHA and AFL-family methods) expose cases where either of f or g *changes behavior* more in response to a shared change in an input seed. Intuitively, cases where one implementation’s behavior is much more affected by a change to an input than the other are “on the path” to exposing full behavioral divergence. This requires a new approach to novelty evaluation not previously explored in the fuzzing literature to our knowledge (relative/response-based novelty), where the behavior of a mutated seed is considered *relative to the original seed behavior*.
- Second, the extent of such changes can be used to inform scheduling and energy decisions; the more input changes differently affect behavior, the more the input merits urgent investigation and substantial computation resources.
- Third, these basic ideas can be further extended in the case where f and g share a common code base, since f is a new version of g , not an independent implementation. In these cases program edges can be mapped into sets of shared and differing code locations, and the structure of the code changes can be exploited to better focus on the changes made to g .
- Finally, the fuzzer can identify when certain byte locations (or higher-order structure entities, in theory) or mutation operators are especially prone to produce large divergences in response for f and g and spend more time applying those more-productive mutants.

1.2 First-Class Differential Specification

Lack of fuzzer awareness of differential structure contributing to weakness in finding differentially-defined bugs is almost certainly *not* the primary reason differential testing is not more widely used, even ignoring the cost of developing independent implementations (which may now be mitigated by AI code synthesis). Many developers do not use differential testing methods because in practice, the problem is seldom one of comparing two implementations with identical behavior, and the effort required to make differential testing actually useful seems overwhelming. While the first prong of this proposal will, we believe, greatly improve the effectiveness of differential testing for those already using it, it cannot be widely used without advances in differential specification.

Differential testing harnesses such as that for the widely used Pyfakefs Python mock filesystem [12] often devote up to half their code to allowance for divergences between a tested and reference system. Moreover, the intellectual effort required to construct such abstraction code and determine which divergences are important and which are spurious is a major burden on would-be users of differential testing, even when analysis reveals the categorization to be, in some sense, “obvious.”

Gulzar et al. in 2019 reported on the (large and growing, at that time) adoption of differential testing at Google [13].

2 Intellectual Merit

2.1 PI Qualifications

PI Groce has a long history with software analysis and verification, including involvement in design and development of well-known fuzzing and testing tools, including the TSTL testing tool for Python, and the DeepState unit-fuzzing interface for C++ and C code; he is at present lead developer and maintainer for multiple open source testing and fuzzing tools, with over 100 GitHub stars (and, in one case, more than 800 stars); he has, in the past, managed releases of industrial open source testing tools with more than 3000 GitHub stars. Groce typically reports 50-100 bugs detected using such tools to open source projects each year, and has been awarded significant bug bounties for security-critical compiler bugs detected using his work. Groce was twice employed by the Bitcoin Core team to investigate the effectiveness of their fuzzing framework, a key component in ensuring the security of the leading implementation of the most important cryptocurrency; this effort included evaluations of the Bitcoin Core fuzzing team’s own internal work on differential fuzzing of Bitcoin versions. His work has consistently involved a marriage of theory of testing and analysis with practical work oriented towards real-world developers, implemented in working tools (all, as will be the case with this project, released as open source). Groce is author of one of the most-cited papers on differential testing [3] (the 7th most cited paper on the topic, according to Google scholar) and the TSTL testing DSL for Python was the first (and, to our knowledge, thus far *only*) such language to include automation of differential harness construction, including complex mappings on inputs and outputs, as a first-class construct in declarative testing [14, 15].

3 Research Plan

3.1 Products

Primary products of this proposal are therefore:

1. An implementation of differential-aware fuzzing algorithms as a new mode of operation for the AFL++ fuzzer, including novel seed scheduling, energy, and mutation approaches based on methods; these will include methods that generalized to all differential fuzzing problems and to the specialized case of versions with largely overlapping implementations.
2. Modalities for automating common cases where the task is differential testing of two binaries that operate over a file or standard input, the common use of AFL and other fuzzers: these will include calibration tools for taking binaries and automatically proposing a differential comparison scheme (composed of those available to users) that is sufficiently sensitive to detect subtle bugs, but applies simple abstractions and choices of detection channels to reduce obvious false positives.

3. A framework/light DSL for declarative, legible, differential harness construction that works with the above fuzzing algorithms and cooperates with the DeepState unit-fuzzing framework, but does not require use of DeepState.
4. Evaluations of the above; these will be performed both in terms of standard fuzzer evaluation methods, using existing differential harnesses and problems that have been addressed in previous work on differential fuzzing/testing and in terms of LLM-based case studies, exploring both the ability of LLMs to make use of the differential testing framework/DSL and the effectiveness of differential fuzzing in agentic coding workflows (including ability to prompt for tailored, highly divergent, implementations).

4 Related Work

Previous work on differential testing is considerable, but previous work on *general mechanisms* for either finding differences in versions or specifying differential comparison problems is relatively limited.

Differential testing (and fuzzing) for specialized settings: McKeeman’s paper on using differential testing to find bugs in compilers is the earliest and most influential work in the field [2]. Compiler testing (as well as other language tool testing, e.g., for debuggers [16] or static analysis tools [17]) has a long and distinguished line of work based in differential approaches [18, 19, 20]; indeed, by and large differential testing is the only means for finding wrong-compile bugs that is in widespread use, and is essential to the success of tools such as Csmith [21]. File systems have also been a target for differential approaches [3]. Recently machine learning systems have also been a focus of differential approaches [22, 23]. Security and privacy testing also makes significant use of differential approaches [24, 25].

Differential testing and fuzzing as a first-class problem: As noted, the most important such work, with respect to detection of differential bugs, is probably that of Pesios et al. on the NEZHA framework [6]. As discussed extensively above, this work revitalized interest in the problem of working out *general purpose* methods of finding divergences between software systems using state-of-the-art fuzzers. While the core algorithmic fuzzing advance presented may be less powerful than the NEZHA work estimated, due to the likelihood that AFL/AFL++ methods for novelty detection are similar in power to the proposed approaches, the general framework for posing the problem and focusing on executions that can reveal differences is a major precursor to this proposal’s approach.

PI Groce’s previous work in the TSTL testing framework for Python is, to our knowledge, the most extensive previous effort to make differential specification an easier problem, by mapping sequences of actions over two implementations into each other, and allowing for use of abstraction functions over inputs and outputs in such a setting [14, 15].

5 Broader Impacts

Improving Software System Reliability: A key element of this proposal’s approach is to focus on realistically deployable techniques. All methods developed will aim for seamless integration with the AFL++ fuzzing system, and thus immediate practical applicability to real-world problems. Differential testing is widely used in various aspects of critical systems infrastructure, and improvements in the ability to detect bugs using differential methods are likely to contribute to software reliability. Additionally, as noted in the problem statement, it seems likely that differential approaches are both made more practical by, and essential for the reliability of, LLM and agentic coding efforts. It is likely that even future workflows with few humans in the loop will make use of differential techniques, and given that fuzzing driven by random mutations is likely an orthogonal source of bugs to LLM-based static analysis of code, such methods will be more important for finding bugs made by LLMs that they cannot reliably detect by only their own “native” methods.

Education and Outreach: The proposed research yields several opportunities for enhancing CS education, recruiting new CS majors, and retaining CS students. PI Groce will work with the NAU Student ACM Chapter to present a series of “excursions in testing” that introduce automated testing to students, using differential

approaches to find bugs in real world code, including code from media player libraries. The work of Guzdial [26] has shown that media computation is an effective way to both recruit and retain students in computer science. Groce also teaches classes (undergraduate and graduate, including a required course in NAU’s Software Engineering major) on automated testing that can present project results. Groce’s prior engagement in teaching and mentoring high school students (including as summer students who have published their work in major venues) and working with local technically-oriented 4H chapters, including in rural communities (such as Williams, AZ) surrounding Flagstaff will also allow preparation and presentation of versions of these excursions tailored to high school students’ level of knowledge.

6 Results From Prior NSF Support

The most relevant prior NSF support for PI Groce is CCF-2129446, “Feedback-Driven Mutation Testing for Any Language,” with a total budget of \$500,000 from 9/2021 until 8/2024, a collaborative proposal with Claire Le Goues of Carnegie Mellon University. *Intellectual Merit:* This project focuses on allowing developers to improve testing by using mutation testing to identify weaknesses in tests. This project has resulted in six publications [10, 4, 27, 28, 29, 30] thus far and two major software releases. *Broader Impact:* Work from this project has already resulted in the reporting and correction of multiple bugs in software systems, including widely used production compilers. The focus on practical discovery of software bugs and compiler testing is a direct path to the proposed project’s approach.

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